

# AICE3001/AICE6002: Mathematics of Machine Learning

## Subsidiary Notes — Lecture 3: Taylor's Expansion

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## 1 Keywords

- Taylor expansion, convergence, remainder, gradient, tangent plane, Hessian, minima, maxima, saddle points

## 2 Main Points

### 2.1 Polynomial Approximation

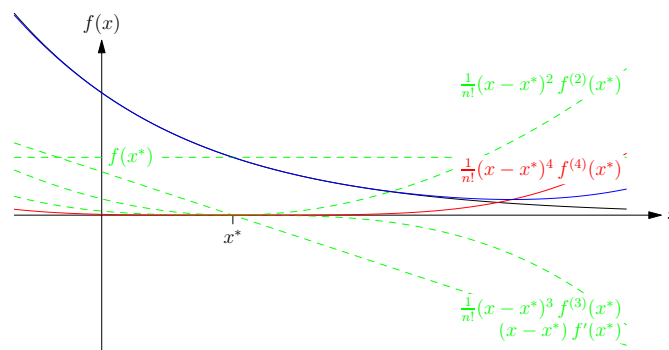
- Much of machine learning is about approximating functions, and every approximation scheme assumes something about how regular the function is
- A Taylor expansion approximates a smooth function near a point  $x^*$  by a polynomial

$$f(x) \approx a_0 + a_1(x - x^*) + a_2(x - x^*)^2 + \cdots + a_n(x - x^*)^n$$

- Differentiating both sides  $i$  times and setting  $x = x^*$  picks out a single coefficient, so if we demand that the two sides agree in value and in their first  $n$  derivatives at  $x^*$  then  $a_i = f^{(i)}(x^*)/i!$
- This gives the Taylor expansion

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \cdots + \frac{1}{n!}f^{(n)}(x^*)(x - x^*)^n$$

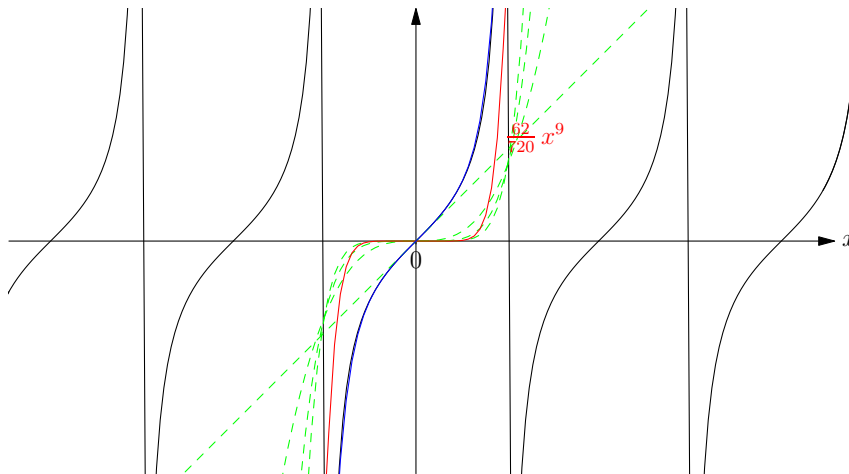
- The figure shows successive Taylor approximations to  $e^x$  around  $x^* = 0$ ; each extra term extends the region where the approximation is good



### 2.2 Convergence

- Adding terms improves the approximation close to  $x^*$ , provided the higher derivatives exist
- Polynomials only diverge as  $x \rightarrow \pm\infty$ , so no polynomial can follow a function that blows up at a finite point, such as  $1/x$ ,  $1/(x-5)^2$  or  $\tan(x)$

- For such functions the Taylor series only converges in a limited region. The lecture animation builds up the expansion of  $\tan(x)$  term by term; the final frame is below. However many terms we add, the series fails beyond the singularities at  $x = \pm\pi/2$

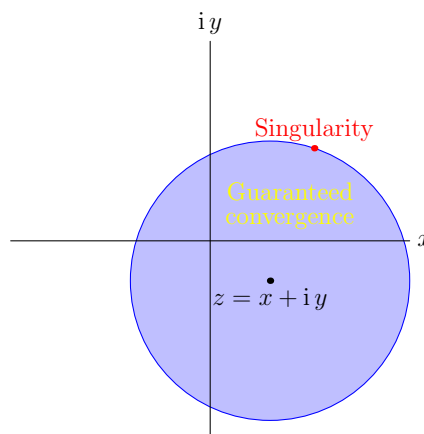


- To understand where a Taylor series converges we need to extend the function into the complex plane,  $z = x + iy$
- A function is *analytic* where its derivative

$$\frac{df(z)}{dz} = \lim_{\epsilon \rightarrow 0} \frac{f(z + \epsilon) - f(z)}{\epsilon}$$

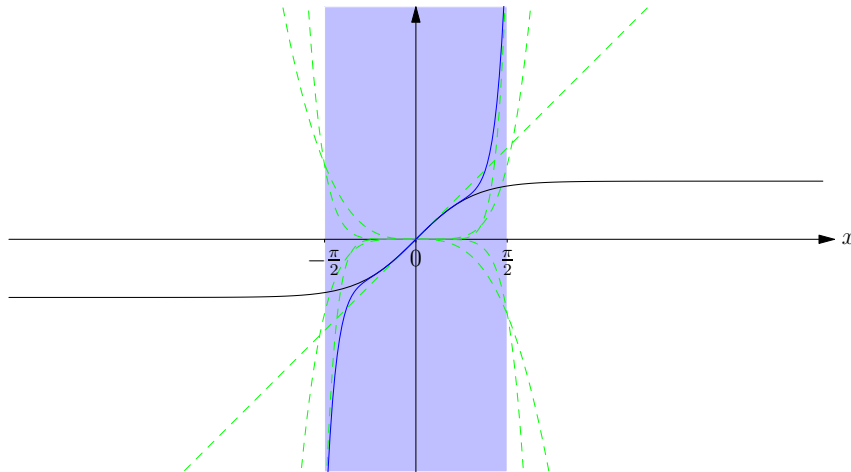
is the same whichever direction the complex number  $\epsilon$  approaches zero from

- polynomials, exponentials, trigonometric functions, logarithms, ratios of polynomials and most "special functions" are analytic except at isolated singular points
- **Key result:** the Taylor series of an analytic function about  $z^*$  converges inside the largest circle centred on  $z^*$  that contains no singularity. The radius of that circle is the *radius of convergence*



- Singularities off the real axis matter even when the function looks perfectly well behaved for real  $x$ 
  - $f(z) = 1/(z + i)^2$  is smooth for every real  $x$ , but it is singular at  $z = -i$ , so its Taylor series about 0 only converges for  $|x| < 1$
- **The hyperbolic tangent.**  $\tanh(x)$  has no singularities on the real line. But  $\tan(ix) = i \tanh(x)$ , and  $\tan$  is singular at  $\pi/2 + k\pi$ , so  $\tanh$  is singular at  $z = \pm i\pi/2, \pm 3i\pi/2, \dots$  on the imaginary axis

- The expansion of  $\tanh(x)$  about 0 therefore converges only for  $|x| < \pi/2$ , even though  $\tanh$  itself is bounded everywhere



## 2.3 The Remainder

- A Taylor expansion comes with an exact error term. Stopping after  $n$  terms,

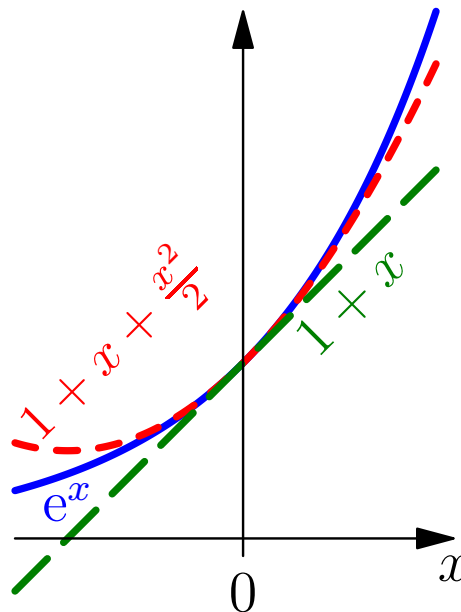
$$f(x) = \sum_{i=0}^{n-1} \frac{(x - x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x - x^*)^n}{n!} f^{(n)}(\theta_x)$$

where  $\theta_x$  is some number between  $x^*$  and  $x$

- This last term, the *remainder*, is very useful for proving inequalities, because we often know the sign of  $f^{(n)}$  even when we don't know  $\theta_x$
- Expanding  $e^x$  about 0,

$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x}$$

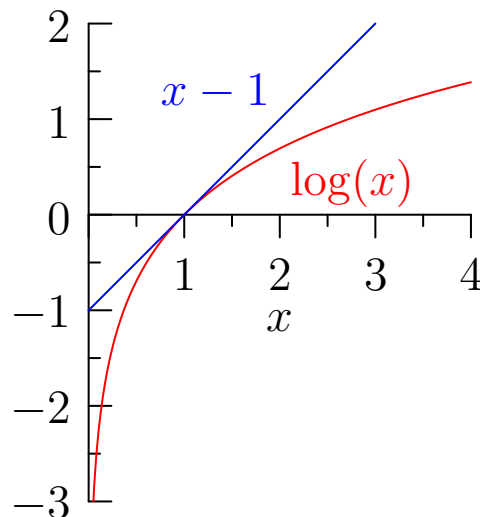
and since  $e^{\theta_x} > 0$  we get the much-used bound  $e^x \geq 1 + x$  for all  $x$



- Expanding  $\log(x)$  about  $x = 1$ ,

$$\log(x) = (x - 1) - \frac{(x - 1)^2}{2\theta^2}$$

so  $\log(x) \leq x - 1$ , or equivalently  $\log(1 + x) \leq x$



- The proof of the remainder formula comes from integrating  $f^{(n)}$  repeatedly; it was skipped in the lecture

## 2.4 High Dimensional Taylor Expansion

- For  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  we approximate by a multivariate polynomial and fix the coefficients by matching derivatives at  $\mathbf{x}^*$  in the same way. To second order,

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*)$$

- $\mathbf{H}$  is the *Hessian*, the matrix of second derivatives

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

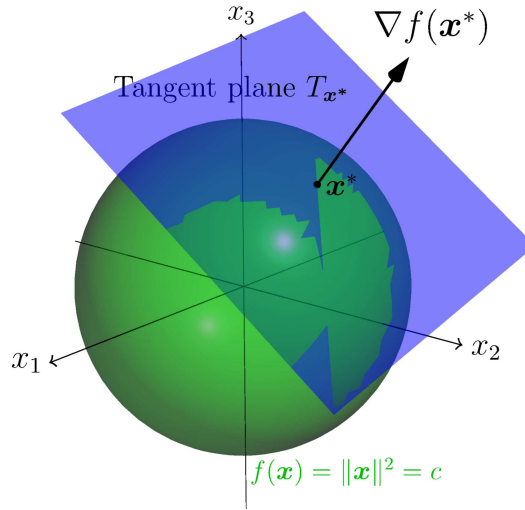
- For functions with continuous second derivatives the order of differentiation does not matter, so the Hessian is symmetric,  $\mathbf{H} = \mathbf{H}^\top$
- Do not confuse the Hessian (second derivatives of a scalar function) with the Jacobian (first derivatives of a vector-valued function)

## 2.5 The Tangent Plane and Steepest Ascent

- For a smooth function the set  $f(\mathbf{x}) = c$  is an  $(n-1)$ -dimensional surface, the high dimensional analogue of a contour line
- The *tangent plane* at  $\mathbf{x}^*$  is the plane that touches this surface at  $\mathbf{x}^*$ . Moving within it changes  $f$  only at second order, so it is the set of points with

$$(\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) = 0$$

that is, the tangent plane is orthogonal to the gradient



- Moving a small distance  $\epsilon$  in a unit direction  $\mathbf{u}$ ,

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}^*) + O(\epsilon^2)$$

- By the Cauchy-Schwarz inequality  $\mathbf{u}^\top \nabla f \leq \|\nabla f\|$ , with equality when  $\mathbf{u}$  points along  $\nabla f$ . So
  - $\nabla f(\mathbf{x}^*)$  is the direction of *steepest ascent*
  - $-\nabla f(\mathbf{x}^*)$  is the direction of *steepest descent*
- This is only a local statement: it holds while the  $O(\epsilon^2)$  term is negligible

## 2.6 The Hessian at Stationary Points

- At a *stationary point*,  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ , the first order term vanishes and

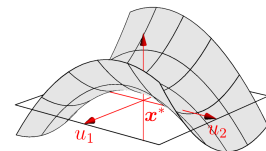
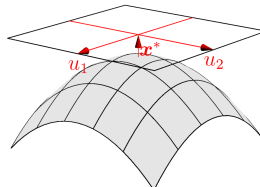
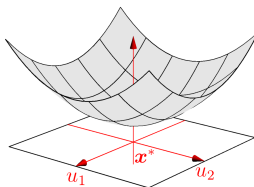
$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \frac{1}{2} \epsilon^2 \mathbf{u}^\top \mathbf{H} \mathbf{u} + O(\epsilon^3)$$

so the sign of  $\mathbf{u}^\top \mathbf{H} \mathbf{u}$  decides whether  $f$  goes up or down

- $\mathbf{H}$  is real and symmetric, so it has real eigenvalues  $\lambda_i$  and orthonormal eigenvectors  $\mathbf{v}_i$  ( $\mathbf{H} \mathbf{v}_i = \lambda_i \mathbf{v}_i$ ,  $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$ )
- Writing any direction as  $\mathbf{u} = \sum_i c_i \mathbf{v}_i$ ,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} = \left( \sum_i c_i \mathbf{v}_i^\top \right) \left( \sum_j c_j \lambda_j \mathbf{v}_j \right) = \sum_{i,j} \lambda_j c_i c_j \mathbf{v}_i^\top \mathbf{v}_j = \sum_i \lambda_i c_i^2$$

- Hence
  - all  $\lambda_i > 0$ :  $\mathbf{H}$  is *positive definite*,  $\mathbf{H} \succ 0$ , and  $\mathbf{x}^*$  is a local minimum
  - all  $\lambda_i < 0$ :  $\mathbf{H}$  is *negative definite*,  $\mathbf{H} \prec 0$ , and  $\mathbf{x}^*$  is a local maximum
  - eigenvalues of both signs:  $\mathbf{x}^*$  is a saddle point
  - if some  $\lambda_i = 0$  the second order test is inconclusive and higher order terms decide



## 2.7 Finding Optima

- To minimise a loss  $L(\mathbf{w})$  we look for  $\nabla L(\mathbf{w}^*) = \mathbf{0}$  with a positive definite Hessian
- Except for very simple losses there are many local minima, and  $\nabla L = \mathbf{0}$  cannot be solved analytically
- Instead we iterate, using gradient descent

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - r \nabla L(\mathbf{w}^{(t)})$$

which moves in the locally steepest downhill direction. Steps shrink as the gradient shrinks near a minimum. Choosing  $r$  well is subtle; we return to this in the optimisation lecture

## 3 Exercises

### 3.1 Radius of convergence

- Write down the Taylor expansion of  $\log(1+x)$  about  $x=0$  up to the  $x^4$  term. Without doing any further calculation, for what values of  $x$  must the series converge?

### 3.2 A bound from the remainder

- Use the remainder form of Taylor's expansion to show that  $\cos(x) \geq 1 - x^2/2$  for all real  $x$

### 3.3 Classifying a stationary point

- For  $f(x, y) = x^2 + 3xy + y^2$  find the gradient and the Hessian. Show that the origin is the only stationary point and classify it

### 3.4 Quadratic functions

- For  $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$  with  $\mathbf{A}$  symmetric, expand  $f(\mathbf{x} + \epsilon \mathbf{u})$  exactly. Hence find the gradient and Hessian, and explain why the second order Taylor expansion of  $f$  is exact

## 4 Answers

### 4.1 Radius of convergence

- The derivatives are  $f^{(k)}(x) = (-1)^{k-1}(k-1)!/(1+x)^k$ , so

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

- $\log(1+z)$  is singular at  $z = -1$ , a distance 1 from the expansion point, so the series converges for  $|x| < 1$

### 4.2 A bound from the remainder

- With  $f(x) = \cos(x)$  we have  $f(0) = 1$ ,  $f'(0) = 0$  and  $f''(\theta) = -\cos(\theta)$ . Stopping after two terms,

$$\cos(x) = 1 + 0 \cdot x - \frac{x^2}{2} \cos(\theta_x)$$

- Since  $\cos(\theta_x) \leq 1$  and  $x^2 \geq 0$ , it follows that  $\cos(x) \geq 1 - x^2/2$

### 4.3 Classifying a stationary point

- $\nabla f = (2x + 3y, 3x + 2y)^\top$ . Setting this to zero gives a linear system with determinant  $2 \times 2 - 3 \times 3 = -5 \neq 0$ , so the only solution is  $x = y = 0$
- The Hessian is constant,

$$\mathbf{H} = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$$

with eigenvalues  $2 \pm 3$ , i.e. 5 and  $-1$ , and eigenvectors  $(1, 1)/\sqrt{2}$  and  $(1, -1)/\sqrt{2}$

- The eigenvalues have opposite signs, so the origin is a saddle point:  $f$  increases along  $(1, 1)$  and decreases along  $(1, -1)$

### 4.4 Quadratic functions

- Expanding,

$$\begin{aligned} f(\mathbf{x} + \epsilon \mathbf{u}) &= \frac{1}{2}(\mathbf{x} + \epsilon \mathbf{u})^\top \mathbf{A} (\mathbf{x} + \epsilon \mathbf{u}) - \mathbf{b}^\top (\mathbf{x} + \epsilon \mathbf{u}) \\ &= f(\mathbf{x}) + \epsilon \mathbf{u}^\top (\mathbf{A} \mathbf{x} - \mathbf{b}) + \frac{1}{2} \epsilon^2 \mathbf{u}^\top \mathbf{A} \mathbf{u} \end{aligned}$$

using  $\mathbf{x}^\top \mathbf{A} \mathbf{u} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{x} = \mathbf{u}^\top \mathbf{A} \mathbf{x}$  for symmetric  $\mathbf{A}$

- Comparing with the Taylor expansion,  $\nabla f = \mathbf{A} \mathbf{x} - \mathbf{b}$  and  $\mathbf{H} = \mathbf{A}$
- There are no terms beyond  $\epsilon^2$ , so the second order expansion is exact. This is why quadratics are such a useful model of a loss function near its minimum